6CS012 – Worksheet 4

Building a Fully Connected Neural Network for Devnagari Handwritten Digit Classification

Tasks covered: Data Preparation → Build FCN → Compile → Train → Evaluate → Save/Load → Predict

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Task 1: Data Preparation

Load images from the dataset folder using PIL, normalize pixel values, and one-hot encode labels.

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

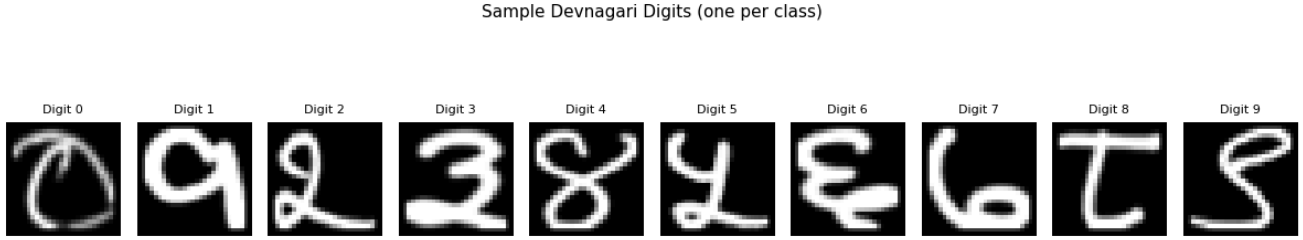
```
import zipfile, os, shutil

# Path to your zip in Google Drive
zip_path = "/content/drive/MyDrive/ Ai and machine learning/Copy of devnagari digit.zip"

# Extract to current Colab directory
with zipfile.ZipFile(zip_path, 'r') as z:
    z.extractall("/content/")

print("Extracted successfully!")
print(os.listdir("/content/DevanagariHandwrittenDigitDataset"))
```

```
Extracted successfully!
['Train', 'Test']
```

```
import os
import numpy as np
import tensorflow as tf
from tensorflow.keras.utils import to_categorical
import matplotlib.pyplot as plt
from PIL import Image

# ── Dataset paths ──────────────────────────────────────────────────────
# Update these paths if your dataset is stored somewhere else.
# If running on Google Colab, upload the zip, unzip it, then set:
#   train_dir = "DevanagariHandwrittenDigitDataset/Train/"
#   test_dir  = "DevanagariHandwrittenDigitDataset/Test/"
train_dir = "/content/DevanagariHandwrittenDigitDataset/Train/"
test_dir  = "/content/DevanagariHandwrittenDigitDataset/Test/"

# ── Image dimensions ───────────────────────────────────────────────────
IMG_HEIGHT, IMG_WIDTH = 28, 28
NUM_CLASSES = 10

# ── Helper: load all images from a folder ──────────────────────────────
def load_images_from_folder(folder):
    """Load images + labels from a directory with sub-folders per class."""
    images, labels = [], []

    # Sort so that digit_0 → 0, digit_1 → 1, …, digit_9 → 9
    class_names = sorted(os.listdir(folder))
    class_map   = {name: idx for idx, name in enumerate(class_names)}

    for class_name in class_names:
        class_path = os.path.join(folder, class_name)
        if not os.path.isdir(class_path):
            continue                        # skip stray files
        label = class_map[class_name]

        for filename in os.listdir(class_path):
            img_path = os.path.join(class_path, filename)
            img = Image.open(img_path).convert("L")             # grayscale
            img = img.resize((IMG_WIDTH, IMG_HEIGHT))           # 28×28
```

```
            img = np.array(img) / 255.0                         # normalise [0,1]
            images.append(img)
            labels.append(label)

    return np.array(images), np.array(labels)

print("Loading training images …")
x_train_raw, y_train_raw = load_images_from_folder(train_dir)

print("Loading test images …")
x_test_raw,  y_test_raw  = load_images_from_folder(test_dir)

# ── Flatten for FCN: (N, 28, 28) → (N, 784) ──────────────────────────────
x_train = x_train_raw.reshape(-1, IMG_HEIGHT * IMG_WIDTH)
x_test  = x_test_raw.reshape( -1, IMG_HEIGHT * IMG_WIDTH)

# ── One-hot encode labels ────────────────────────────────────────────────
y_train = to_categorical(y_train_raw, num_classes=NUM_CLASSES)
y_test  = to_categorical(y_test_raw,  num_classes=NUM_CLASSES)

print(f"\nTraining set : {x_train.shape}  |  Labels: {y_train.shape}")
print(f"Test set     : {x_test.shape}   |  Labels: {y_test.shape}")

# ── Visualise a sample of 10 training images ─────────────────────────────
plt.figure(figsize=(12, 3))
for i in range(10):
    plt.subplot(1, 10, i + 1)
    plt.imshow(x_train_raw[i * 1700], cmap="gray")   # one image per class
    plt.title(f"Digit {i}", fontsize=8)
    plt.axis("off")
plt.suptitle("Sample Devnagari Digits (one per class)", fontsize=11)
plt.tight_layout()
plt.show()
```

```
Loading training images …
Loading test images …

Training set : (17000, 784)  |  Labels: (17000, 10)
Test set     : (3000, 784)   |  Labels: (3000, 10)
```




```
import os

# NOTE: Replace 'YOUR_DATASET_ZIP_URL' with the actual URL of your dataset zip file.
# For example, if it's hosted on Google Drive, you'd use a direct download link.
# If you've uploaded it directly, you can skip this step and just use the filename.
zip_file_url = "YOUR_DATASET_ZIP_URL" # Example: "https://github.com/some_user/some_repo/raw/main/DevanagariHandwrittenDigitDatas
zip_file_name = "DevanagariHandwrittenDigitDataset.zip"

# If you are providing a URL, uncomment the line below:
# !wget -O {zip_file_name} {zip_file_url}

# If you have uploaded the zip file manually to the Colab environment, just ensure it's in the current directory.

print(f"Checking for {zip_file_name}...")
if os.path.exists(zip_file_name):
    print(f"{zip_file_name} found.")
else:
    print(f"WARNING: {zip_file_name} not found. Please upload it or provide a valid download URL.")
```

```
Checking for DevanagariHandwrittenDigitDataset.zip...
WARNING: DevanagariHandwrittenDigitDataset.zip not found. Please upload it or provide a valid download URL.
```

```
import os

zip_file_name = "DevanagariHandwrittenDigitDataset.zip"
```

```
destination_folder = "DevanagariHandwrittenDigitDataset"

if os.path.exists(zip_file_name):
    print(f"Unzipping {zip_file_name}...")
    !unzip -q {zip_file_name} -d {destination_folder}
    print(f"Dataset unzipped to {destination_folder}/")

    # Verify contents
    if os.path.exists(os.path.join(destination_folder, 'Train')):
        print("Train directory found within unzipped dataset.")
    else:
        print("WARNING: Train directory not found in the unzipped dataset. Check the zip file structure.")

    if os.path.exists(os.path.join(destination_folder, 'Test')):
        print("Test directory found within unzipped dataset.")
    else:
        print("WARNING: Test directory not found in the unzipped dataset. Check the zip file structure.")

else:
    print(f"Skipping unzip: {zip_file_name} not found. Please ensure the zip file is present.")
```

```
Skipping unzip: DevanagariHandwrittenDigitDataset.zip not found. Please ensure the zip file is present.
```

```
# Update these paths if the unzipped folder has a different name or structure.
train_dir = "DevanagariHandwrittenDigitDataset/Train/"
test_dir  = "DevanagariHandwrittenDigitDataset/Test/"

print(f"train_dir is set to: {train_dir}")
print(f"test_dir is set to: {test_dir}")

# You can now re-run the 'Task 1: Data Preparation' cell.
```

```
train_dir is set to: DevanagariHandwrittenDigitDataset/Train/
test_dir is set to: DevanagariHandwrittenDigitDataset/Test/
```

Task 2: Build the FCN Model

Sequential model with 3 hidden layers (64 → 128 → 256 neurons, sigmoid activation) and a softmax output layer.

```
 from tensorflow import keras
from tensorflow.keras import layers

# ── Model definition ──────────────────────────────────────────────────────
model = keras.Sequential(
    [
        keras.Input(shape=(IMG_HEIGHT * IMG_WIDTH,)),   # 784-dim input vector

        # Hidden layer 1
        layers.Dense(64,  activation="sigmoid", name="hidden_1"),

        # Hidden layer 2
        layers.Dense(128, activation="sigmoid", name="hidden_2"),

        # Hidden layer 3
        layers.Dense(256, activation="sigmoid", name="hidden_3"),

        # Output layer – 10 classes, probabilities via softmax
        layers.Dense(NUM_CLASSES, activation="softmax", name="output"),
    ],
    name="Devnagari_FCN"
)

model.summary()
```

```
Model: "Devnagari_FCN"
```

Layer (type)	Output Shape	Param #
hidden_1 (Dense)	(None, 64)	50,240
hidden_2 (Dense)	(None, 128)	8,320
hidden_3 (Dense)	(None, 256)	33,024
output (Dense)	(None, 10)	2,570

```
Total params: 94,154 (367.79 KB)
Trainable params: 94,154 (367.79 KB)
Non-trainable params: 0 (0.00 B)
```

Task 3: Compile the Model

Use Adam optimiser, categorical cross-entropy loss (labels are one-hot encoded), and accuracy metric.

```
model.compile(
    optimizer="adam",
    loss="categorical_crossentropy",   # one-hot encoded labels → categorical
    metrics=["accuracy"]
)

print("Model compiled successfully!")
print(f"  Optimizer : adam")
print(f"  Loss      : categorical_crossentropy")
print(f"  Metrics   : accuracy")
```

```
Model compiled successfully!
  Optimizer : adam
  Loss      : categorical_crossentropy
  Metrics   : accuracy
```

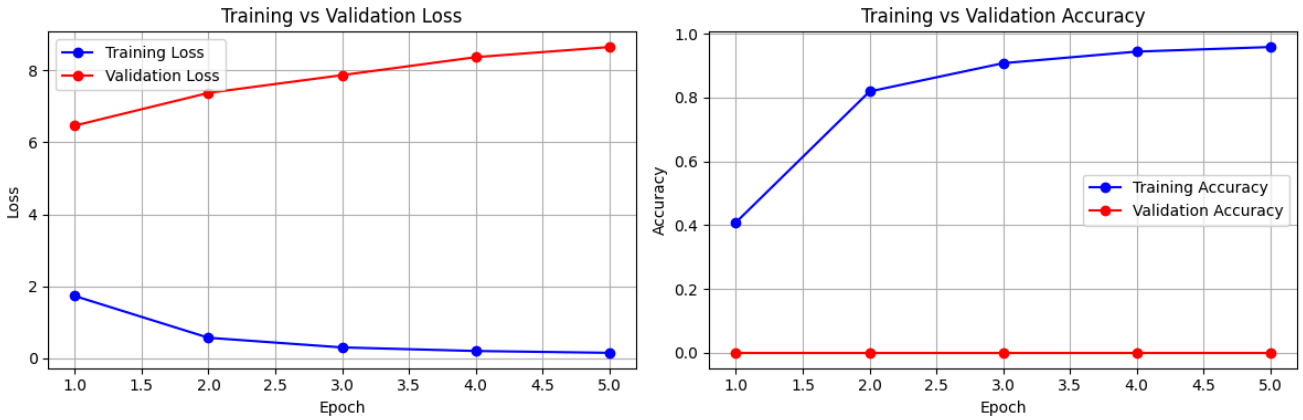
Task 4: Train the Model

Train for up to 20 epochs (batch size = 128, 20 % validation split).
Callbacks: **ModelCheckpoint** saves the best weights; **EarlyStopping** halts training if val_loss stops improving.

```
BATCH_SIZE = 128
EPOCHS     = 20

# ── Callbacks ──────────────────────────────────────────────────────────
callbacks = [
    keras.callbacks.ModelCheckpoint(
        filepath="best_devnagari_model.keras",   # saves the BEST model (lowest val_loss)
        monitor="val_loss",
        save_best_only=True,
        verbose=1
    ),
    keras.callbacks.EarlyStopping(
        monitor="val_loss",
        patience=4,                              # stop after 4 epochs with no improvement
        restore_best_weights=True,
        verbose=1
    ),
]

# ── Train ──────────────────────────────────────────────────────────────
history = model.fit(
    x_train,
    y_train,
    batch_size=BATCH_SIZE,
    epochs=EPOCHS,
    validation_split=0.2,     # 20% of training data used for validation
    callbacks=callbacks,
    verbose=1
)

print("\nTraining complete!")
```

```
Epoch 1/20
107/107 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.2252 - loss: 2.0715
Epoch 1: val_loss improved from None to 6.46713, saving model to best_devnagari_model.keras

Epoch 1: finished saving model to best_devnagari_model.keras
107/107 ━━━━━━━━━━━━━━━━━━━━ 3s 9ms/step - accuracy: 0.4087 - loss: 1.7260 - val_accuracy: 0.0000e+00 - val_loss: 6.4671
Epoch 2/20
 97/107 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.7869 - loss: 0.7148
Epoch 2: val_loss did not improve from 6.46713
107/107 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - accuracy: 0.8191 - loss: 0.5653 - val_accuracy: 0.0000e+00 - val_loss: 7.3757
Epoch 3/20
102/107 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.8887 - loss: 0.3349
Epoch 3: val_loss did not improve from 6.46713
107/107 ━━━━━━━━━━━━━━━━━━━━ 1s 7ms/step - accuracy: 0.9079 - loss: 0.2968 - val_accuracy: 0.0000e+00 - val_loss: 7.8698
Epoch 4/20
 97/107 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9412 - loss: 0.2105
Epoch 4: val_loss did not improve from 6.46713
107/107 ━━━━━━━━━━━━━━━━━━━━ 1s 7ms/step - accuracy: 0.9442 - loss: 0.1974 - val_accuracy: 0.0000e+00 - val_loss: 8.3700
Epoch 5/20
107/107 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9568 - loss: 0.1513
Epoch 5: val_loss did not improve from 6.46713
107/107 ━━━━━━━━━━━━━━━━━━━━ 1s 7ms/step - accuracy: 0.9585 - loss: 0.1462 - val_accuracy: 0.0000e+00 - val_loss: 8.6519
Epoch 5: early stopping
Restoring model weights from the end of the best epoch: 1.

Training complete!
```

```
# ── Visualise training progress ──────────────────────────────────────────
train_loss = history.history["loss"]
val_loss   = history.history["val_loss"]
train_acc  = history.history["accuracy"]
val_acc    = history.history["val_accuracy"]
epochs_ran = range(1, len(train_loss) + 1)

plt.figure(figsize=(12, 4))

# Loss curve
plt.subplot(1, 2, 1)
plt.plot(epochs_ran, train_loss, "b-o", label="Training Loss")
plt.plot(epochs_ran, val_loss,   "r-o", label="Validation Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Training vs Validation Loss")
plt.legend()
plt.grid(True)

# Accuracy curve
plt.subplot(1, 2, 2)
plt.plot(epochs_ran, train_acc, "b-o", label="Training Accuracy")
plt.plot(epochs_ran, val_acc,   "r-o", label="Validation Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.title("Training vs Validation Accuracy")
plt.legend()
plt.grid(True)

plt.tight_layout()
plt.savefig("training_curves.png", dpi=150)
plt.show()
print("Training curves saved to training_curves.png")
```




```
Training curves saved to training_curves.png
```

Task 5: Evaluate the Model

Measure the final loss and accuracy on the held-out **test set**.

```
test_loss, test_acc = model.evaluate(x_test, y_test, verbose=2)

print(f"\n{'='*40}")
print(f"  Test Loss     : {test_loss:.4f}")
print(f"  Test Accuracy : {test_acc * 100:.2f} %")
print(f"{'='*40}")
```

```
94/94 - 0s - 3ms/step - accuracy: 0.6007 - loss: 2.0333

========================================
  Test Loss     : 2.0333
  Test Accuracy : 60.07 %
========================================
```

Task 6: Save and Load the Model

Persist the trained model to disk in `.h5` format, reload it, and verify it still achieves the same accuracy.

```
# ── Save ──────────────────────────────────────────────────────────
MODEL_PATH = "devnagari_fcn_model.h5"
model.save(MODEL_PATH)
print(f"Model saved → {MODEL_PATH}")
```

```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format
Model saved → devnagari_fcn_model.h5
```

```
# ── Load and re-evaluate ──────────────────────────────────────────
loaded_model = tf.keras.models.load_model(MODEL_PATH)
print(f"Model loaded from {MODEL_PATH}")

loaded_loss, loaded_acc = loaded_model.evaluate(x_test, y_test, verbose=2)
print(f"\nLoaded model – Test Accuracy: {loaded_acc * 100:.2f} %")
```

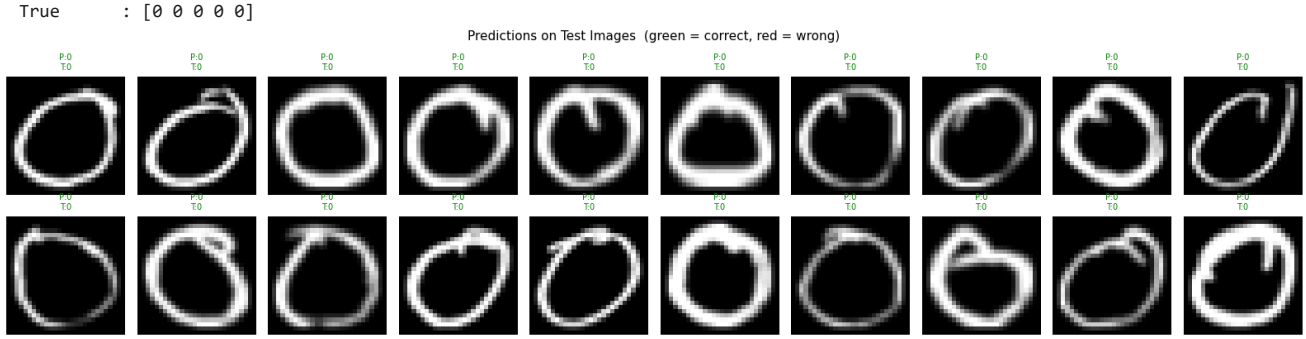
```
WARNING:absl:Compiled the loaded model, but the compiled metrics have yet to be built. `model.compile_metrics` will be empty unt
Model loaded from devnagari_fcn_model.h5
94/94 - 1s - 6ms/step - accuracy: 0.6007 - loss: 2.0333

Loaded model – Test Accuracy: 60.07 %
```

Task 7: Make Predictions

Run the model on the test set, convert softmax probabilities to digit labels, and display a grid of results.

```
# ── Predict ──────────────────────────────────────────────────────────
predictions      = loaded_model.predict(x_test, verbose=0)
predicted_labels = np.argmax(predictions, axis=1)   # highest-probability class
true_labels      = np.argmax(y_test, axis=1)

# Quick sanity check on the first 5 samples
print("Sample predictions (first 5 test images):")
print(f"  Predicted : {predicted_labels[:5]}")
print(f"  True      : {true_labels[:5]}")

# ── Visualise 20 predictions ─────────────────────────────────────────
plt.figure(figsize=(16, 4))
for i in range(20):
    plt.subplot(2, 10, i + 1)
    plt.imshow(x_test_raw[i], cmap="gray")
    colour = "green" if predicted_labels[i] == true_labels[i] else "red"
    plt.title(f"P:{predicted_labels[i]}\nT:{true_labels[i]}",
              fontsize=7, color=colour)
    plt.axis("off")

plt.suptitle("Predictions on Test Images  (green = correct, red = wrong)",
             fontsize=11)
plt.tight_layout()
plt.savefig("predictions_grid.png", dpi=150)
plt.show()
print("Prediction grid saved to predictions_grid.png")
```

```
Sample predictions (first 5 test images):
  Predicted : [0 0 0 0 0]
  True      : [0 0 0 0 0]
```




```
Prediction grid saved to predictions_grid.png
```

Summary

Step	Detail
Dataset	Devnagari Handwritten Digits – 17 000 train / 3 000 test
Input	28 × 28 grayscale → flattened to 784-dim vector
Architecture	Dense(64, sigmoid) → Dense(128, sigmoid) → Dense(256, sigmoid) → Dense(10, softmax)
Optimiser	Adam